

Network Scan Project

Security Assessment Report

Local Network Reconnaissance & Packet Analysis using Nmap and Wireshark

Author	B.R. Harshitha
Email	harshithabrreddy@gmail.com
LinkedIn	linkedin.com/in/b-r-harshitha-17769b383/
GitHub	github.com/harshithabrreddy/Network-Scan-Project
Institution	Gates Institute of Technology
Program	B.Tech in Cyber Security (2022–2026)
Date	May 2026
Classification	Academic / Lab Project – Confidential

1. Objective

The objective of this project is to perform local network scanning using Nmap and analyze captured network packets using Wireshark. The exercise simulates real-world reconnaissance tasks performed by SOC Analysts and Security Engineers to understand network topology, identify active hosts, detect running services, and observe packet-level behavior on a controlled lab environment.

2. Lab Environment & Tools

2.1 Environment Setup

Platform	VMware Workstation (Virtualized Lab)
Operating System	Kali Linux (Attacker/Scanner VM)
Target Network	192.168.93.0/24 (Internal Virtual Network)
Target Hosts	Multiple Windows and Linux virtual machines

Capture Tool	Wireshark (Live Packet Capture – .pcapng format)
--------------	--

2.2 Tools Used

- Kali Linux – Attacker machine and primary scanning OS
- Nmap (Network Mapper) – Port scanning, service detection, OS fingerprinting
- Wireshark – Packet capture and traffic analysis
- VMware Workstation – Virtualization platform for isolated lab

3. Methodology

3.1 Phase 1 – Host Discovery (TCP SYN Scan)

A TCP SYN Stealth Scan was performed to identify all active hosts on the 192.168.93.0/24 subnet. This scan sends SYN packets to each host and analyzes responses to determine which hosts are online without completing the full TCP handshake.

```
sudo nmap -sS 192.168.93.0/24
```

Key flags used:

- -sS : TCP SYN scan (stealth scan, does not complete 3-way handshake)
- 192.168.93.0/24 : Target subnet range (all 254 possible hosts)

3.2 Phase 2 – Service & OS Detection

A service version and OS fingerprinting scan was conducted to enumerate running services and identify operating systems on discovered hosts.

```
sudo nmap -sV -O 192.168.93.0/24
```

Key flags used:

- -sV : Version detection (probes open ports to identify service/version)
- -O : OS detection (uses TCP/IP stack fingerprinting to guess OS)

3.3 Phase 3 – Packet Capture with Wireshark

Wireshark was used simultaneously to capture all network traffic generated during the Nmap scans. The resulting .pcapng file was analyzed to observe TCP SYN packets, ARP requests, acknowledgements, and retransmissions to understand network behavior at the packet level.

4. Key Findings

4.1 Summary of Findings

#	Finding	Risk Level	Recommendation
F-01	Multiple active hosts identified on 192.168.93.0/24	Info	Maintain accurate asset inventory
F-02	Apache HTTP Server detected on Port 80	High	Upgrade to HTTPS (Port 443), disable HTTP
F-03	Open ports discovered on multiple hosts	Medium	Close unused ports, apply firewall rules
F-04	Windows and Linux OS identified via fingerprinting	Medium	Keep OS patched and updated
F-05	TCP retransmissions observed in Wireshark	Low	Monitor network for stability issues
F-06	ARP broadcast traffic captured	Info	Implement Dynamic ARP Inspection (DAI)

4.2 Network Activity Observed in Wireshark

- TCP SYN packets – Outbound from scanner to each host port (scan initiation)
- TCP SYN/ACK packets – Received from open ports on target hosts
- TCP RST packets – Returned by closed ports (port rejection)
- ARP Request/Reply – Address resolution for active hosts on the subnet
- TCP Retransmission – Observed on some hosts indicating network instability

5. Security Risk Analysis

5.1 Open HTTP Service (Port 80)

An Apache HTTP server was detected listening on Port 80. HTTP traffic is transmitted in plaintext, meaning any data exchanged (including credentials or session tokens) can be intercepted via a man-in-the-middle (MITM) attack.

Risk Level: HIGH

- All web services should be migrated to HTTPS (TLS/SSL) on Port 443
- HTTP traffic should be redirected to HTTPS using 301 permanent redirects
- HTTP Strict Transport Security (HSTS) header should be enabled

5.2 Open Ports & Increased Attack Surface

Multiple open ports were discovered across hosts on the network. Each open port represents a potential entry point for an attacker if the associated service is misconfigured, outdated, or unpatched.

Risk Level: MEDIUM

- Apply the principle of least privilege – close all ports not required for operations
- Implement firewall rules (iptables / Windows Firewall) to restrict access
- Regularly audit open ports using network scanning tools

5.3 OS Detection via Fingerprinting

Operating system types were successfully identified through Nmap OS fingerprinting. This information can be leveraged by attackers to identify OS-specific vulnerabilities.

Risk Level: MEDIUM

- Apply network-level obfuscation to reduce OS fingerprint accuracy
- Keep all OS patches and security updates current
- Monitor for unusual scanning activity in SIEM/IDS logs

6. Scan Results Overview

results.txt – Initial Nmap Scan Output

The initial scan output (results.txt) captured the list of active hosts, open ports, and basic service banner information. Key observations from this file included host IP addresses, port states (open/closed/filtered), and initial service guesses based on well-known port numbers.

final_results.txt – Detailed Service & OS Scan

The final_results.txt file contained enriched output from the -sV -O scan, including confirmed service names, version numbers (e.g., Apache httpd version), and Nmap OS guess confidence scores for each detected host.

scan_capture.pcapng – Wireshark Packet Capture

The Wireshark capture file documented all network-level activity during the scans. Analysis of this file revealed the TCP handshake process, ARP activity, scan packet patterns, and retransmission events, providing a complete network-layer view of the reconnaissance process.

7. Conclusion

This project successfully demonstrated the process of local network reconnaissance using industry-standard tools in a controlled virtual lab environment. Key learning outcomes include:

- Understanding network topology and host discovery via TCP SYN scanning
- Service enumeration and OS fingerprinting using Nmap version/OS detection flags
- Packet-level analysis using Wireshark including TCP, ARP, and retransmission events
- Identification of real security risks such as open HTTP services and exposed ports
- Documentation of findings and security recommendations in a structured report format

The skills developed in this project directly apply to SOC Analyst, Vulnerability Assessment, and Penetration Testing roles, particularly in the areas of network reconnaissance, threat identification, and security reporting.

8. References & Tools

- Nmap Official Documentation – <https://nmap.org/docs.html>
- Wireshark User Guide – <https://www.wireshark.org/docs>
- OWASP Transport Layer Protection Cheat Sheet
- NIST SP 800-115 – Technical Guide to Information Security Testing
- Kali Linux Documentation – <https://www.kali.org/docs>